

3.2 Moment generating function and density

3.2.1 Moment generating functions of random vectors

The moment generating function of a random vector $\mathbf{Y} \in \mathbb{R}^n$ is $M_{\mathbf{Y}}(\mathbf{t}) = \mathbb{E} \exp(\mathbf{t}^\top \mathbf{Y})$, and Theorem 2.6.1 proves the two properties we need when it is finite on a neighbourhood of $\mathbf{0}$. *Uniqueness*: two random vectors whose moment generating functions are finite and equal near $\mathbf{0}$ have the same distribution. *Factorization*: two blocks $\mathbf{Y}_1, \mathbf{Y}_2$ are independent iff $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1)M_{\mathbf{Y}_2}(\mathbf{t}_2)$ near $\mathbf{0}$. The second extends to k blocks by induction. If $M_{\mathbf{Y}}$ is the product of the k block functions, setting $\mathbf{t}_1 = \mathbf{0}$ shows that $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$ has the product of the last $k-1$, so $\mathbf{Y}_1$ is independent of $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$, whose blocks are mutually independent by the induction hypothesis. The converse is immediate.

Theorem 3.2.1 — Moment generating function

A random vector $\mathbf{Y}$ is $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ iff

$$M_{\mathbf{Y}}(\mathbf{t}) = \exp(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) \quad \text{for every } \mathbf{t} \in \mathbb{R}^n. \quad (3.2.1)$$

Consequently the distribution of a normal vector is determined by its mean vector and covariance matrix.

Proof. Suppose $\mathbf{Y}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. By Eq. 3.1.1, $\mathbf{t}^\top \mathbf{Y} \sim N(\mathbf{t}^\top \boldsymbol{\mu}, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$, and $M_{\mathbf{Y}}(\mathbf{t})$ is the moment generating function of this scalar evaluated at $s = 1$. Lemma 3.1.1(a) gives Eq. 3.2.1.

Conversely, suppose Eq. 3.2.1 holds for a symmetric $\boldsymbol{\Sigma}$. Fix $\mathbf{a}$. The moment generating function of $\mathbf{a}^\top \mathbf{Y}$ at s is $M_{\mathbf{Y}}(s\mathbf{a}) = \exp\{(\mathbf{a}^\top \boldsymbol{\mu})s + \frac{1}{2}(\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})s^2\}$. The coefficient $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$ is nonnegative, since otherwise this function would tend to zero as $s \rightarrow \pm\infty$, whereas every random variable X has $\mathbb{E} e^{sX} + \mathbb{E} e^{-sX} \geq \mathbb{E} e^{|sX|} \geq 1$. By Lemma 3.1.1(b), $\mathbf{a}^\top \mathbf{Y} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. So $\mathbf{Y}$ is normal, and matching means and variances of all linear combinations shows $\mathbb{E} \mathbf{Y} = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$.

Finally, two normal vectors with the same $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ have the same moment generating function Eq. 3.2.1, which is finite everywhere, so they have the same distribution by the uniqueness property. ■

The last sentence is what makes $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ a well-defined distribution. Combined with Theorem 3.1.2, it says that $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ holds iff $\mathbf{Y}$ has the same distribution as $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$ for some, and then every, matrix $\mathbf{A}$ with $\mathbf{A}\mathbf{A}^\top = \boldsymbol{\Sigma}$. The number of columns of $\mathbf{A}$, and so the length of $\mathbf{Z}$, does not matter. Some books take this representation as the definition (Christensen 2020; Seber and Lee 2003). The proof above shows the two definitions agree.

The moment generating function also encodes moments. Differentiating Eq. 3.2.1 at $\mathbf{t} = \mathbf{0}$ gives $\partial M_{\mathbf{Y}}/\partial \mathbf{t} = \boldsymbol{\mu}$ and $\partial^2 M_{\mathbf{Y}}/\partial \mathbf{t} \partial \mathbf{t}^\top = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top = \mathbf{E}(\mathbf{Y}\mathbf{Y}^\top)$. Higher derivatives give the fourth moments needed for variances of quadratic forms in Chapter 4 (Exercise (B1)).

3.2.2 The density

Theorem 3.2.2 — Density

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- (a) If $\boldsymbol{\Sigma}$ is positive definite, $\mathbf{Y}$ has the density

$$f(\mathbf{y}) = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} \exp\left\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})\right\}, \quad \mathbf{y} \in \mathbb{R}^n. \quad (3.2.2)$$

- (b) Conversely, a random vector with density Eq. 3.2.2, for some positive definite $\boldsymbol{\Sigma}$, is $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
- (c) If $\boldsymbol{\Sigma}$ is singular, $\mathbf{Y}$ has no density on $\mathbb{R}^n$.

Proof.

- (a) By Theorem 3.2.1, $\mathbf{Y}$ has the same distribution as $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$ with $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$, so it suffices to find the density of the latter. The entries of $\mathbf{Z}$ are independent standard normals, so $\mathbf{Z}$ has density $\prod_i (2\pi)^{-1/2} e^{-z_i^2/2} = (2\pi)^{-n/2} \exp(-\frac{1}{2}\mathbf{z}^\top \mathbf{z})$. Since $\boldsymbol{\Sigma}$ is positive definite, so is $\boldsymbol{\Sigma}^{1/2}$ (its eigenvalues are the positive square roots of those of $\boldsymbol{\Sigma}$), and the map $\mathbf{z} \mapsto \mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{z}$ is an affine bijection of $\mathbb{R}^n$ with inverse $\mathbf{z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{y} - \boldsymbol{\mu})$. The change-of-variables formula multiplies the density of $\mathbf{Z}$ at $\mathbf{z}(\mathbf{y})$ by $|\det \boldsymbol{\Sigma}^{-1/2}| = \det(\boldsymbol{\Sigma})^{-1/2}$. Finally $\mathbf{z}^\top \mathbf{z} = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma}^{-1/2} (\mathbf{y} - \boldsymbol{\mu}) = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})$.
- (b) A density determines a distribution, and by (a) the $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution has this density.
- (c) $\mathbf{Y}$ lies with probability one in $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$, an affine subspace of dimension less than n . It has Lebesgue measure zero, so it would receive probability zero under any distribution with a density. ■

A singular normal vector does have a density with respect to the natural measure on the affine subspace where it lives (Proposition 3.3.4 and Exercise (B2)), but that is rarely needed. In this book, arguments are made with the moment generating function or with linear transformations, which work in every case, and densities are used only for pictures and likelihoods.

Example. (The bivariate normal)

For $n = 2$, write $\sigma_1^2, \sigma_2^2 > 0$ for the variances and ρ for the correlation. Then

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \quad \det \boldsymbol{\Sigma} = \sigma_1^2 \sigma_2^2 (1 - \rho^2),$$

so Σ is positive definite iff $|\rho| < 1$. With standardized coordinates $u_i = (y_i - \mu_i)/\sigma_i$, the density Eq. 3.2.2 is

$$f(y_1, y_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{u_1^2 - 2\rho u_1 u_2 + u_2^2}{2(1-\rho^2)}\right\}.$$

When $\rho = 0$ it factors into the two univariate normal densities. When $\rho = \pm 1$ there is no density: $\text{Var}(u_2 \mp u_1) = 0$ in standardized units, so all the probability lies on a line.

3.2.3 Contours and the Mahalanobis distance

For positive definite Σ , the density Eq. 3.2.2 depends on $\mathbf{y}$ only through

$$\Delta^2(\mathbf{y}) = (\mathbf{y} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu}),$$

the squared Mahalanobis distance from $\mathbf{y}$ to $\boldsymbol{\mu}$ introduced in Section 2.2. The density is a decreasing function of Δ^2 , so its contours are the ellipsoids $\{\mathbf{y} : \Delta^2(\mathbf{y}) = c^2\}$. Their shape comes from the spectral decomposition $\Sigma = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$ (Theorem 1.7.2). In the coordinates $w_i = \mathbf{u}_i^\top(\mathbf{y} - \boldsymbol{\mu})$ we have $\Delta^2 = \sum_i w_i^2 / \lambda_i$, so the ellipsoid has principal axes along the eigenvectors $\mathbf{u}_i$, with half-lengths $c\sqrt{\lambda_i}$. Large-variance directions are long axes. The Mahalanobis distance measures distance from the centre in units of standard deviation along each principal direction. The volume of $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq c^2\}$ is proportional to $\prod_i \sqrt{\lambda_i} = \det(\Sigma)^{1/2}$, so the generalized variance $\det \Sigma$ (Section 2.1) measures the size of the region the distribution occupies. The factor $\det(\Sigma)^{-1/2}$ in Eq. 3.2.2 spreads the same total probability over that region: a larger generalized variance means a flatter density.

Proposition 3.2.3

If $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ with Σ positive definite, then $\Delta^2(\mathbf{Y})$ has the chi-squared distribution with n degrees of freedom, $\chi^2(n)$, which is $\chi^2(n, 0)$ in the notation of Definition 4.1.1. So the ellipsoid $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$, where q is the $1 - \alpha$ quantile of $\chi^2(n)$, has probability exactly $1 - \alpha$.

Proof. Put $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$. For every $\mathbf{a}$, $\mathbf{a}^\top \mathbf{Z} = (\Sigma^{-1/2}\mathbf{a})^\top \mathbf{Y} - \mathbf{a}^\top \Sigma^{-1/2}\boldsymbol{\mu}$ is a linear combination of $\mathbf{Y}$ minus a constant, hence normal by Definition 3.1.1, so $\mathbf{Z}$ is normal. Its mean is zero and its covariance is $\mathbf{I}$ (Proposition 2.2.4(a)). By Theorem 3.2.1 its moment generating function is $\exp(\frac{1}{2}\mathbf{t}^\top \mathbf{t}) = \prod_i \exp(\frac{1}{2}t_i^2)$, which factorizes, so its entries are independent standard normals. $\Delta^2(\mathbf{Y}) = \mathbf{Z}^\top \mathbf{Z} = \sum_i Z_i^2$ is a sum of n independent squared standard normals, which is the definition of $\chi^2(n)$. ■

Chapter 4 generalizes this considerably, to $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ with a nonzero mean and a general $\mathbf{A}$ (Theorem 4.3.2). Because the density is largest inside these ellipsoids, they are the smallest regions with their probability content, which is why they reappear as confidence regions for $\boldsymbol{\beta}$ in Chapter 12.

Computing with the density. Neither Σ^{-1} nor $\det \Sigma$ should be formed explicitly. With the Cholesky factorization $\Sigma = \mathbf{L}\mathbf{L}^\top$, we have $\Delta^2(\mathbf{y}) = \|\mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu})\|^2$, computed by a triangular solve, and $\log \det \Sigma = 2 \sum_i \log l_{ii}$. The listing evaluates the log density this way, checks it

against `scipy`, and counts how many of the simulated vectors fall inside the 95% ellipsoid. The fraction is 0.9500. The average of Δ^2 over the draws is 3.004 and its variance is 5.994, against the $\chi^2(3)$ values 3 and 6.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])

N = 400_000
a = np.array([0.3, -1.0, 2.0])
q = [0.05, 0.25, 0.5, 0.75, 0.95]
exact = stats.norm.ppf(q, loc=a @ mu, scale=np.sqrt(a @ Sigma @ a))
cov_err = max(np.abs(np.cov(Y, rowvar=False) - Sigma).max() for Y in (Y_chol, Y_root))

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)                # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T       # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))

def normal_logpdf(y, mu, Sigma):
    """log density of N(mu, Sigma) at the rows of y, Sigma positive definite."""
    L = np.linalg.cholesky(Sigma)
    w = np.linalg.solve(L, (y - mu).T)       # w = L^{-1}(y - mu), so w^T w =
    Delta^2
    D2 = np.sum(w**2, axis=0)
    logdet = 2.0 * np.sum(np.log(np.diag(L)))
    return -0.5 * (len(mu) * np.log(2 * np.pi) + logdet + D2), D2

logf, D2 = normal_logpdf(Y_chol, mu, Sigma)
inside = np.mean(D2 <= stats.chi2.ppf(0.95, df=3))
print(f"fraction inside the 95% ellipsoid: {inside:.4f}")
```

[Run this code online](#)

3.2.4 Exercises

3.2.4.1 A. Check your understanding

Exercise A1.

Show that $f(y_1, y_2) = k \exp\{-\frac{1}{2}(2y_1^2 + 2y_1y_2 + y_2^2 - 4y_1 - 2y_2)\}$ is a bivariate normal density for a suitable constant k . Find k , μ , Σ and the correlation.

Solution. The quadratic part is $\mathbf{y}^\top \mathbf{P} \mathbf{y}$ with $\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, which is positive definite with $\det \mathbf{P} = 1$. Matching $(\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{y}^\top \mathbf{P} \mathbf{y} - 2\boldsymbol{\mu}^\top \mathbf{P} \mathbf{y} + \boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu}$ with the linear terms requires $\mathbf{P} \boldsymbol{\mu} = (2, 1)^\top$, so $\boldsymbol{\mu} = (1, 0)^\top$ and $\boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu} = 2$. The exponent is $-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) + 1$. By Theorem 3.2.2(b) this is the $N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ density with $\boldsymbol{\Sigma} = \mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ provided $ke = (2\pi)^{-1}(\det \boldsymbol{\Sigma})^{-1/2} = (2\pi)^{-1}$. So $k = e^{-1}/(2\pi)$, and the correlation is $-1/\sqrt{2}$.

Exercise A2.

Let $\boldsymbol{\Sigma}$ have unit variances and correlation ρ , with $|\rho| < 1$. Find the principal axes and half-lengths of the ellipse $\Delta^2 = c^2$, show that its area is $\pi c^2 \sqrt{1 - \rho^2}$, and show that it contains probability $1 - e^{-c^2/2}$.

3.2.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{X} = \mathbf{Y} - \boldsymbol{\mu}$. Expand the moment generating function of $\mathbf{X}$ in a power series to show that all odd-order moments of $\mathbf{X}$ vanish and

$$E(X_i X_j X_k X_l) = \sigma_{ij} \sigma_{kl} + \sigma_{ik} \sigma_{jl} + \sigma_{il} \sigma_{jk}.$$

Deduce that $\text{Var}(\mathbf{X}^\top \mathbf{A} \mathbf{X}) = 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma})$ for symmetric $\mathbf{A}$.

Solution. By Theorem 3.2.1, $M_{\mathbf{X}}(\mathbf{t}) = \exp(\frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) = \sum_{m \geq 0} (\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^m / (2^m m!)$, a series containing only terms of even degree in $\mathbf{t}$. Mixed moments of order d are the d th partial derivatives at $\mathbf{0}$, which come from the degree- d terms, so odd-order moments vanish. The degree-four term is $p(\mathbf{t}) = \frac{1}{8} (\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^2 = T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$, where $T(\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}) = \frac{1}{8} (\mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{v})(\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{x})$ is 4-linear. Differentiating $T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$ once in each of the directions $\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k, \mathbf{e}_l$ gives the sum of T over the 24 ways of placing these four vectors in the four slots. This holds whether or not the indices are distinct, since it is the product rule applied to a multilinear form. A placement contributes $\frac{1}{8} \sigma_{ab} \sigma_{cd}$, where the first two slots receive the indices a, b and the last two receive c, d . Each of the three ways of splitting the four *directions* into two pairs, $\{ij, kl\}, \{ik, jl\}, \{il, jk\}$, arises from 8 placements (choose which pair goes first, and the order within each pair), which gives the formula. Then, for symmetric $\mathbf{A}$,

$$E(\mathbf{X}^\top \mathbf{A} \mathbf{X})^2 = \sum_{i,j,k,l} a_{ij} a_{kl} E(X_i X_j X_k X_l) = (\text{tr} \mathbf{A} \boldsymbol{\Sigma})^2 + 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}),$$

and subtracting $(E \mathbf{X}^\top \mathbf{A} \mathbf{X})^2 = (\text{tr} \mathbf{A} \boldsymbol{\Sigma})^2$ gives the variance.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\text{rank}(\boldsymbol{\Sigma}) = r < n$, and use the notation of Proposition 3.3.4. Show that $\mathbf{V} = \mathbf{Q}^\top (\mathbf{Y} - \boldsymbol{\mu})$ has a density on $\mathbb{R}^r$, and that this density can be written as $(2\pi)^{-r/2} (\lambda_1 \cdots \lambda_r)^{-1/2} \exp\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^+ (\mathbf{y} - \boldsymbol{\mu})\}$ with $\mathbf{y} = \boldsymbol{\mu} + \mathbf{Q} \mathbf{v}$, where $\boldsymbol{\Sigma}^+ = \mathbf{Q} \boldsymbol{\Lambda}^{-1} \mathbf{Q}^\top$ is the Moore–Penrose inverse.

Exercise B3.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ positive definite and let $\boldsymbol{\nu}$ be a fixed vector. Show that $(\mathbf{Y} - \boldsymbol{\nu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\nu})$ has the noncentral chi-squared distribution $\chi^2(n, \gamma)$ with $\gamma = (\boldsymbol{\mu} - \boldsymbol{\nu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \boldsymbol{\nu})$, in the convention of Definition 4.1.1, and find its mean.

3.2.4.3 C. Going deeper**Exercise C1.**

Let $\mathbf{Y}$ have density Eq. 3.2.2, and let $E = \{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$. Show that every Borel set A with $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$ has volume at least that of E . *Hint:* compare $\int_{A \setminus E} f$ with $\int_{E \setminus A} f$.

Solution. The density is a decreasing function of Δ^2 , so with $c = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} e^{-q/2} > 0$ we have $f \geq c$ on E and $f < c$ off E . We may assume A has finite volume. Subtracting $\Pr(\mathbf{Y} \in A \cap E)$ from both sides of $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$ gives $\int_{A \setminus E} f \geq \int_{E \setminus A} f$. Hence

$$c \operatorname{vol}(A \setminus E) \geq \int_{A \setminus E} f \geq \int_{E \setminus A} f \geq c \operatorname{vol}(E \setminus A),$$

so $\operatorname{vol}(A \setminus E) \geq \operatorname{vol}(E \setminus A)$. Adding $\operatorname{vol}(A \cap E)$ to both sides gives $\operatorname{vol}(A) \geq \operatorname{vol}(E)$. The argument uses only that E is a set on which f is at least as large as anywhere outside it, so it applies to the level sets of any density.